N83 first-fall certificates

Balanced candidate domains · m = 2–9 · 27 October 2026

32 exact first-fall values; 16 lower bounds; 48 certificate replays passed. Every Koblitz and general-binary input has first fall 2. Every prime input has a certified lower bound of 5. The exact prime values remain unresolved.

N83 first fall: exact values and lower bounds

Planted / natural targets; all saved input equations checked

	Koblitz	Binary	Prime
m=2	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)
m=3	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)
m=4	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)
m=5	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)
m=6	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)
m=7	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)
m=8	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)
m=9	2 / 2	2 / 2	>=5 (bound) / >=5 (bound)

Green: attained degree with explicit witness. Amber: exclusion bound; exact value unresolved.

Source: evidence/results.json. Unit: total polynomial degree. No timing or statistical claim.

Workload. All 48 original synthetic inputs: three curve families, eight arities, one planted and one natural target. ell = floor(83/m): 41, 27, 20, 16, 13, 11, 10, 9. Candidate x-domain size is 2^ell; points still require lifting to the curve. No target resampling or outcome selection.

Binary certificate. The original equations contain no nonzero generator of degree below 2. A saved constant combination of the 83 quadratic equations gives a nonzero linear polynomial. Quadratic-part rank is 81 at m=2 and 82 at m=3–9; full rank is 83 throughout. Forward and reverse pivot orders agree. Each combination was reconstructed from the original equations in a fresh process. Thus the earliest new degree fall is 2, also a nontrivial leading-form syzygy at 2.

Prime certificate. Selector equations form a monic degree-two Groebner basis with no degree falls. After quotienting their top squares, a diagonal coefficient minor of the m−1 quartic S3 rows has nonzero determinant: 16 at m=2 and 64 otherwise. This excludes new falls through degree 4. It does not establish a fall at 5.

Limits. These values depend on the specific S3-chain encoding. Last fall and regularity remain unmeasured. The earlier regularity run's 96 resource-limited computations are preserved separately. One target per kind does not support a statistical curve-family claim. Fresh-process replay uses the same implementation and is not an independent research review. No performance or ECDLP recovery claim.

Validation and provenance. Seven native tests passed; positive, duplicate, homogeneous and zero-row controls passed. All 48 certificate replays and the raw file hashes passed. The unified export validates with file verification. Source, binary, input list and individual systems are pinned in `protocol.json`. EKS job `n83-first-fall-m2-m9-20261027-r3` exited 0; two CPUs, 8 GiB, 900-second cap, no retries. Wall times are diagnostic.

Reproduce and inspect. `evidence/results.json` holds every polynomial witness or coefficient minor; `evidence/replay.jsonl` records replays; `experiment.export.json` binds the protocol and raw outputs. The graph is generated from those records by `../src/bin/fall_report.rs`. Full definitions and execution rules are in `DESIGN.md`.

Primary references. Kosters and Yeo: trace identities Caminata and Gorla: first and last fall Kousidis and Wiemers: Boolean filtered algebra.